

KANAGARAJAH THILAKSHAN

DATA SCIENCE ENGINEER

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PROFESSIONAL SUMMARY

Data Science undergraduate at SLIIT with a strong foundation in programming, data analysis, machine learning, artificial intelligence, natural language processing (NLP), statistical analysis, exploratory data analysis (EDA), predictive modelling, and full-stack web development. Experienced in developing academic and practical projects using Python, Flask, Django, FastAPI, Java, SQL, MongoDB, React, and machine learning libraries including Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn. Skilled in data preprocessing, feature engineering, data visualization, model evaluation, and text classification to derive actionable insights and support data-driven decision making. Brings professional experience in UI/UX development and pharmacy operations, combining analytical thinking, problem-solving, accurate record management, user-focused design, and collaborative teamwork to deliver efficient and impactful technical solutions.

WORK EXPERIENCE

UI/UX Developer (Part-Time) | Nov 2024 - Jun 2025

- Tested user interfaces across desktops, tablets, and mobile devices, identifying layout issues and improving responsive user experience.
- Resolved technical design challenges by applying modern UI/UX principles and collaborating with team members to improve user experience and interface usability.

Assistant Pharmacist - Staff | Kurinchy | Nov 2021 - Present

- Maintained accurate medication records and supported compliance with pharmacy regulations, quality standards, and safe prescription handling practices.
- Assisted junior pharmacy assistants with pharmacy software usage, prescription processing workflows, and daily operational procedures.
- Provided training support to team members on data-related software tools and analysis techniques used for operational tracking and decision support.

PROJECTS

Hall Booking Mobile Application | Mar 2026 - May 2026 [GitHub Repository](#)

Stack: Expo, Express.js, JavaScript, JWT, MongoDB, Node.js, React Native, Stripe API

- Established a cross-platform hall booking and event management application using React Native, Expo, Node.js, and MongoDB, supporting User, Hall Owner, and Admin roles through Role-Based Access Control (RBAC).
- Crafted and implemented secure RESTful APIs with Express.js, integrating JWT authentication, bcrypt password hashing, and middleware-based authorization.
- Architected and optimized MongoDB database schemas for bookings, payments, users, reviews, and notifications using Mongoose ODM to ensure efficient data management.
- Integrated the Stripe Payment Gateway to enable secure online transactions, payment verification, and systematized refund processing.
- Built a real-time booking management system featuring slot availability validation, dynamic pricing calculation, and automated notification delivery.
- Created an administrative dashboard for user management, hall approval workflows, booking oversight, and revenue monitoring.

Machine Learning-Powered Smart Timetable Management System | Feb 2026 - May 2026 [GitHub Repository](#)

Stack: CSS, Flask, HTML, JavaScript, Machine Learning, Pandas, Python

- Developing an intelligent timetable management system to automatically generate optimized schedules for lecturers and students.
- Performed Exploratory Data Analysis (EDA), data preprocessing, and predictive modeling techniques to reduce scheduling conflicts and improve timetable efficiency.
- Built a web interface to support timetable viewing, management, and user-friendly interaction with generated schedules.

Machine Learning-Based Fake Review Detection System | July 2025 - Sep 2025

Stack: Flask, Machine Learning, NLP, Pandas, Python.

- Built a machine learning system to identify deceptive product reviews using text analysis and NLP-based feature extraction.
- Used supervised learning, preprocessing, text classification, and model evaluation metrics such as accuracy, precision, and recall.
- Conducted text-based data analysis, feature engineering, and model evaluation to improve classification accuracy.
- Created a simple Flask-based web interface for real-time fake review prediction and practical demonstration.

Tourism Itinerary Management System | Mar 2025 [GitHub Repository](#)

- Developed a travel planning system that helps users select destinations and create structured itineraries with daily schedules.
- Focused on simple trip management workflows to make destination planning more organized, efficient, and user-friendly.

Book Store Management System | Nov 2023 - Jan 2024

Stack: CSS, HTML, Java, MySQL.

- Designed and improved a bookstore management system for handling book inventory, customer details, and sales records.
- Implemented core features for adding, updating, and searching books with database integration for efficient data storage and retrieval.

Smart Greenhouse Monitoring System | Sep 2023 - Oct 2023

Stack: Arduino, C/C++, Sensors.

- Enhanced an Arduino-based greenhouse monitoring system to track temperature, humidity, and soil moisture in real time.
- Integrated sensors and mechanized actions such as watering and ventilation to support better plant growth conditions.

EDUCATION

BSc (Hons) Information Technology Specializing in Data Science | SLIIT, Northern Uni, Jaffna | Jul 2023 - Present

- Core focus: Data Science, Database Systems, Machine Learning, Natural Language Processing, Programming, Web Technologies.

G.C.E Advanced Level (A/L) | Jaffna Hindu College, Jaffna | Aug 2021

Diploma in IT & Languages (English & Sinhala) | ICBT, Jaffna | Oct 2023

RELEVANT COURSEWORK

- Machine Learning
- Data Mining
- Artificial Intelligence
- Data Structures and Algorithms
- Database Management Systems
- Statistics for Data Science
- Natural Language Processing

CERTIFICATIONS

- AI Poster Participation - Think AI Expo | SLIIT Northern UNI, Presented an AI research poster, 2025
- Diploma in English and Information Technology | ICBT Campus, 2023
- Gateway Foundation Program | John Keels, 2018

TECHNICAL SKILLS

Programming Languages: CSS, HTML, Java, JavaScript, Python, SQL.

Machine Learning & Data Science: Data Analysis, Data Preprocessing, Data Visualization, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation, Natural Language Processing (NLP), Predictive Analytics, Predictive Modelling, Statistical Analysis, Supervised Learning, and Text Classification.

Data Handling & Libraries: Data Cleaning, Data Transformation, Matplotlib, NumPy, Pandas, Scikit-learn, Seaborn, SQL Queries.

Web Development: Django, FastAPI, Flask, React, REST APIs.

Tools & Platforms: GitHub Repository, Jupyter Notebook, MongoDB, PyCharm, VS Code.

Languages: English, Tamil

REFERENCES

Mr. K. Thiruthanigesan (Senior Lecturer)

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